

Hapiko Thermal Printer — Command Set Reference (for CM)

This is the complete set of ESC/POS-style serial commands the firmware sends to the thermal printer. It is intended as a bring-up reference for the contract manufacturer implementing the printer-board command set.

Source of truth: `components/peripherals/thermal_printer.cpp` (driver is an ESP-IDF port of the Adafruit Thermal Printer library). All print paths — image printing, factory test, test pages — go through this one driver, so this list is exhaustive.

Serial interface

Parameter	Value
Interface	UART, 8 data bits, no parity, 1 stop bit, no HW flow control
Baud rate	230400
Print width	384 px max (48 bytes/row)
DTR/handshaking	GPIO input, pulled up; printer drives it LOW = ready
Firmware-version gating	Some commands differ for printer firmware <code>>= 264</code> vs older. Firmware assumes v2.68 by default.

Command table

Bytes shown as hex. `n` = a single parameter byte. Order is exactly what the firmware emits.

POWER / LIFECYCLE

Command	Bytes	Use
Wake byte	FF	Sent on power-on and on wake-from-sleep to wake the controller.
Disable sleep (fw $\geq$ 264)	1B 38 00 00 (ESC 8 0 0)	Keep printer awake; sent at end of wake sequence.
Wake sequence (fw < 264)	ten 00 bytes	Legacy wake for old firmware (NUL padding with delays).
Enter sleep (fw $\geq$ 264)	1B 38 01 00 (ESC 8 1 0)	Low-power sleep after 1 second of idle; sent after each print job.
Enter sleep (fw < 264)	1B 38 01 (ESC 8 1)	Legacy single-byte sleep timeout.
Reset / initialize	1B 40 (ESC @)	Reset settings to default and clear the buffer; sent during init.

PRINT CONFIGURATION (SENT DURING INIT)

Command	Bytes	Use
Set heating params	1B 37 n1 n2 n3 (ESC 7 ...)	Print control: n1 =max heating dots (11), n2 =heat time (120 = 1.2 ms), n3 =heat interval (40 = 400 μ s).
Set line spacing	1B 33 n (ESC 3 n)	Line spacing in dots. Init sets n=30 .
Set character spacing	1B 20 n (ESC SP n)	Right-side char spacing in dots. Only used by the built-in test page.

PRINTING

Command	Bytes	Use
Print text	raw ASCII bytes	Text content (<code>print()</code> / <code>println()</code>).
Line feed	0A (LF)	Print and advance one line; appended by <code>println()</code> .
Tab	09 (HT)	Horizontal tab.

Command	Bytes	Use
Feed lines	1B 64 n (ESC d n)	Advance paper by n text lines. Used by test/factory prints.
Feed dots	1B 4A n (ESC J n)	Advance paper by n dot rows. Production path for top/bottom image margins (fine control).
Print bitmap (raster)	12 2A h w (DC2 * h w) + data	Raster bitmap. h =rows in this chunk (≤ 255), w =bytes per row (≤ 48). Immediately followed by w × h bytes of 1-bpp image data (MSB = leftmost pixel, 1 = black). Tall images are split into multiple DC2 * chunks. This is the main image-print command.

STATUS / HANDSHAKING

Command	Bytes	Use
Enable status-back / DTR	1D 61 20 (GS a 32)	Enable Automatic Status Back so the printer drives the DTR line for hardware flow control; sent during init.
Query paper status (fw ≥ 264)	1B 76 00 (ESC v 0)	Returns 1 status byte. Bit 3 set ⇒ out of paper. (0x60 = paper present, 0x6C = no paper on the current part.)
Query paper status (fw < 264)	1D 72 01 (GS r 1)	Legacy paper/status query.

Defined but NOT sent by current firmware

These exist in the driver API but are never invoked on any runtime path. Flagged so the CM doesn't need to prioritize them, and because their encodings may warrant review:

Command	Bytes	Notes
Set print density	12 23 n (DC2 # n)	n = (density << 5) break_time . Density/break defaults (10/2) are applied via the heat command instead; this command is not sent.
Set max printing dots	1B 4D n (ESC M n)	Defined as setMaxPrintingDots ; never called. (ESC M is non-standard for this purpose — confirm intended encoding before relying on it.)

Key constants / defaults

Setting	Default	Where
Baud rate	230400	DEFAULT_PRINTER_BAUD
Assumed printer firmware	v2.68 (firmware_version = 268)	gates wake/sleep/status command form
Heat time	120 (1.2 ms)	Adafruit standard
Heat interval	40 (400 μs)	Adafruit standard
Max heating dots	11	Adafruit default
Print density / break time	10 / 2	Adafruit default
Max width	384 px (48 bytes)	hardware head width
Max raster chunk height	255 rows	per DC2 * chunk
Feed before image	0.0 lines (configurable)	top margin
Feed after image	4.5 lines (configurable)	bottom margin